
Financial Econometrics

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- 1(a). The table below provides Stata output based on the variable MktRF, a measure of weekly excess US stock market returns (%) from March 2000 to May 2021 ($T = 255$). Using the information provided comment on the statistical properties of MktRF.

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. summarize MktRF, detail
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Mkt-RF				
Percentiles		Smallest		
1%	-10.72	-17.23		
5%	-8.12	-13.38		
10%	-5.99	-10.72	Obs	255
25%	-1.96	-10.35	Sum of wgt.	255
50%	1.17	Largest	Mean	.5940392
			Std. dev.	4.543082
75%	3.33	10.18		
90%	5.65	11.35	Variance	20.63959
95%	7.63	12.47	Skewness	-.5298389
99%	11.35	13.65	Kurtosis	4.035121

- 1(b). Using the Stata output below, perform a two sided hypothesis test at the 99% significance level that weekly excess US stock market returns (MktRF) has zero autocorrelation at lag 2. [We will cover how to generate such tables in the Stata labs. Note that AC refers to a column with the sample autocorrelations at the corresponding lag. We will not cover the PAC stat in this module. The Q and Prob>Q columns we will discuss in a later lecture.]

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. corrgram MktRF
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LAG	AC	PAC	Q	Prob>Q
1	0.0854	0.0854	1.8799	0.1704
2	-0.0649	-0.0735	2.9697	0.2265
3	0.0591	0.0722	3.8788	0.2749

2. The weekly returns of the Apple stock for 1/1/01 to 12/2/01 are
(0.101, 0.050, 0.135, 0.003, 0.0543, -0.072, -0.007, -0.007, -0.009).
Using this data set of size $T = 9$ answer the following (all the below can be done on a calculator, and I would advise you use a calculator in this example).
- Calculate the sample mean and variance. Using these results calculate the sample skewness and kurtosis.
 - What can we infer from the estimated moments in (a) about the properties of the distribution of the sample data.
 - Perform a test that the true mean of apple returns is zero, against the alternative the true mean is non zero at the 95% significance level. [See the my Youtube video on Sampling and Hypothesis testing for more background on this.]
 - Calculate the sample autocorrelation at lag 1 and 2.
 - Perform a 95% two sided hypothesis test that the population correlation at lag 1 is zero (state all the steps, the test statistic and distribution and any assumptions made).
 - Why may we be sceptical of inference from the hypothesis tests in (c),(e).
3. Below is the joint distribution of returns R_t for $t = 1$ and $t = 2$ that take 3 values, 0, 1 or 2 percent.

		R_2		
		0	1	2
R_1	0	0.05	0.1	0.03
	1	0.21	0.11	0.19
	2	0.08	0.15	0.08

- Derive the marginal distributions of R_1 and R_2 .
- Derive the mean and variance of R_1, R_2 .
- Find $Cov(R_1, R_2)$ and $Cov(R_1^2, R_2^2)$.
- Find the conditional distribution of R_1 given $R_2 = 0, 1$ or 2 respectively. .
- Derive the conditional mean and variance of R_1 given each possible value of R_2 .
- Are the variables R_1 and R_2 statistically independent? Provide a verification for your answer.